winkler

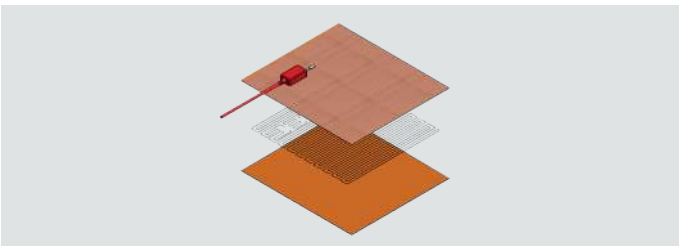
SILICONE HEATING ELEMENTS



Customized silicone heating elements
Silicone heating plates
Silicone heating elements for railway applications

SILICONE HEATING ELEMENTS PREVIEW

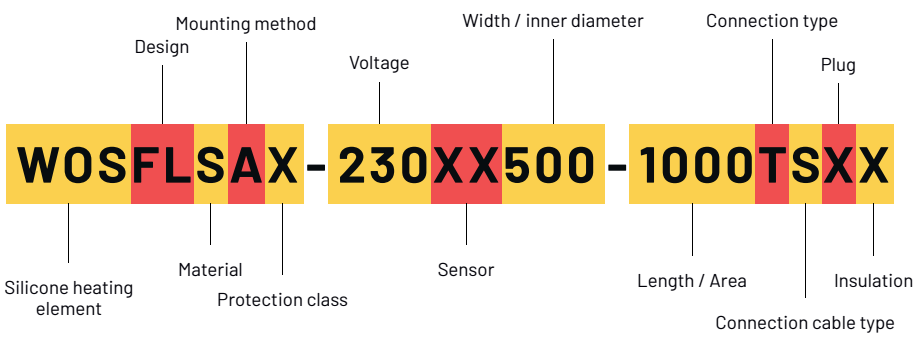
GENERAL PRODUCT DESIGN OF SILICONE HEATING ELEMENTS



Silicone heating elements consist of heating conductor tracks in the form of wire, stranded wire or etched foil, which are embedded between two or more thermally and mechanically resistant silicone layers with additional glass fabric inserts to protect them from moisture and vulcanised to form a homogeneous unit.

EXAMPLES FOR CLASSIFICATION SILICONE HEATING ELEMENTS

All our silicone heating elements have a coded type designation based on the features of the product. If you request a silicone heating element from Winkler you'll receive a type designation as reference that looks like the following example.



*The entire coding is shown on page 26.

WINKLER QUALITY

The technical experience and the high level of quality assurance are reflected in all Winkler products so that our customers can rely on tested and proven products for their applications. Our heating systems are characterized by an even distribution of the heating power and a very generously dimensioned heating conductor arrangement. In this way, we achieve the most direct and gentle heat transfer possible to the medium or the heating object. We only use high-quality, tested and proven materials and components, and there is no compromise in this respect when it comes to the selection of and cooperation with our suppliers. Our customers are therefore offered excellent and trustworthy products with a long service life, even under heavy load conditions. As a matter of principle, Winkler products are 100% routine tested. Even during production, there are documented tests. During the subsequent final inspection, the silicone heating elements are once again subjected to a strict quality inspection. This testing procedure ensures a high degree of safety and reliability. These high quality and safety standards are beneficial. We use an **ISO 9001: 2015** certified quality management system and are a certified manufacturer of products in hazardous areas in accordance with Directive **2014 / 34 / EU Annex VII (ATEX).**



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Cylindrical heating elements
Spiral heating elements
Free form heating elements
Connection options
Control and security options

SILICONE HEATING PLATES 16-19

Product overview
Product examples

SILICONE HEATING ELEMENTS FOR RAILWAY APPLICATIONS 20-25

Railway certification overview EN 45545-2 and NFPA 130
Silicone heating elements
Silicone heating cables

CLASSIFICATION OF SILICONE HEATING ELEMENTS 26

CUSTOMIZED SILICONE HEATING ELEMENTS



USE OF THESE HEATING ELEMENTS IS LIMITED ONLY BY THE IMAGINATION

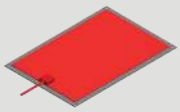
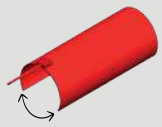
Product overview
Flat heating elements
Cylindrical heating elements
Spiral heating elements
Free form heating elements
Connection options
Control and security options



PRODUCT OVERVIEW SILICONE HEATING ELEMENTS

Select design

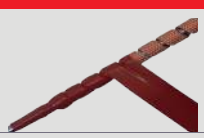
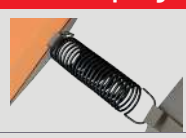
Flat

	❶ Flat	❷ Flat on assembly	❸ Flat and flexible	❹ Cylindrical and flexible	❺ Cyli
					
Flexibility	○	○	Ø > 100 mm	Ø > 20 mm...100 mm	Ø > 20 mm
Min. batch size	●	● ● ●	●	● ● ●	● ●
Protection class					

Select material

	Standard material
	
Models	CE LSLT UL94-V1
Temperature range	-60 °C...180 °C
Robustness / tensile strength	● ● ●
Advantages	■ Smooth surface ■ Slides less ■ Adheres well

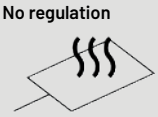
Select mounting

	RTV adhesive	Silicone band	Self-adhesive film	Tension springs
				
Suitable designs	❶ ❷ ❸ ❹ ❺ ❻ ❼ ❽	❶ ❷ ❸ ❹ ❺ ❻ ❼ ❽	❶ ❷ ❸ ❹ ❺ ❻ ❼ ❽	❶ ❷ ❸ ❹ ❺ ❻ ❼ ❽
Heat transfer	● ● ● ● ●	● ● ● ●	● ● ● ●	● ● ●
Removability	○	○	○	● ●
Advantages	■ Universally applicable	■ Self-welding	■ Uniform adhesion	■ High clamping force ■ Self-tensioning

Select options

Control

No regulation



- self-stabilising
- Temperature controlled by heating output and ambient
- Temperature fluctuations possible

Temperature switch



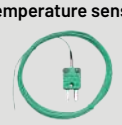
- Breaks the circuit at the target temperature and closes it after cooling

Temperature controller



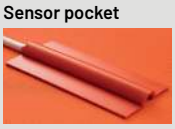
- Precise temperature control
- Temperature measurement using temperature sensors

Temperature sensor



- Accurate temperature measurement
- Thermocouple type K or type J
- Temperature sensor PT100 or PT1000

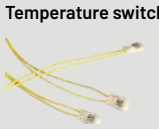
Sensor pocket



- Temperature sensors can be inserted into sensor pockets
- Allows for simple exchange
- Alternatively, temperature sensors can be glued in

Security

Temperature switch



- Automatic shutdown at target temperature
- Multiple use



- Flat design of the temperature switch

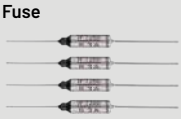


- Temperature switch for lower temperatures
- More precise switching
- Only on request



- Durable design with a high number of switching cycles
- Only on request

Fuse



- Alternative to the temperature switch
- Protection against overheating
- Triggered when the temperature is too high
- One-way fuse

Pl

Stripped with wire end ferrules



- Easy w
- For ap

Round plug



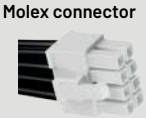
- Plug ar tempe
- Availab

AMP connector



- Conne few ca

Molex connector



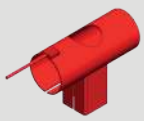
- Standa applica

Harting connector



- For hig elemer
- Wiring possib

Select design

drical	6 Spiralised	Free form	7 Free form	8 Free form on assembly	
ndrical					
...100 mm	Ø 5 mm...50 mm	○		○	Flexibility
● ●	● ● ●	● ● ● ●		● ● ● ●	Min. batch size
					Protection class

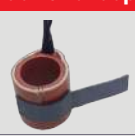
Select material

Extreme material



UL 94-HB/CE	Models
-60 °C...200 °C	Temperature range
● ● ● ● ●	Robustness / tensile strength
■ Structured surface ■ Particularly suitable for flexible designs ■ Slides less ■ Adheres well	Advantages

Select mounting

Eyelets/hooks	Silicone straps	Hook and loop	Snap fasteners	
				
1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8	Suitable designs
● ● ● ●	● ● ●	● ●	● ●	Heat transfer
● ● ●	● ● ● ●	● ● ● ●	● ● ● ● ●	Removability
■ Robust and suitable for complex geometries	■ Self-tensioning	■ Cost-effective and practical	■ Easy assembly and disassembly	Advantages

Select options

ug

viring
plications without connectors

nd play connector with our
rature controllers
ple in 3-, 4-, or 6-pin versions

ctors for applications with
bles

ard connectors for various
ations

gh-performance heating
nts
of multiple cables
le

Connections

Patch



- Flat connector
- Adjustable length and cross-section

Silicone block



- Reduces stress on the connector
- Available in IP64 version

Head with strain relief



- Installation of temperature switches and fuses possible

Soldering pads



- Connection option for individual adjustments

Thermal insulation

Standard insulation foam



- Talc-coated surface
- It is glued to the surface

UL insulation foam



- Textured surface
- UL certified
- Adhesively bonded to the surface

Pipe insulation



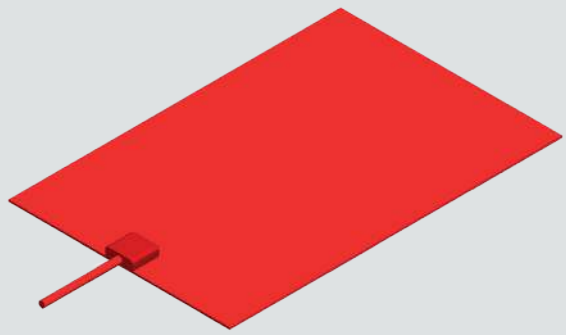
- Quickly snaps into place for easy installation
- Improves the efficiency of coil heaters
- Holds coil heaters in place without any fastening

General technical data

Dimensions	max. B960 mm x L2600 mm
Thickness	min. 1.5 mm
Weight	min. 2 kg/m²
Operating voltage	12-480 V
Heating power	max. 0.5 W/cm²
Ingress protection	max. IP65 (in accordance)

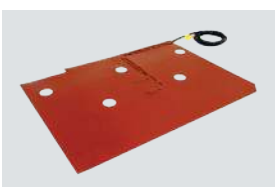
FLAT HEATING ELEMENTS

WOSFL



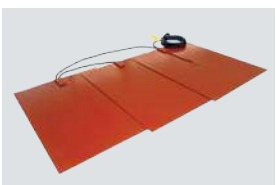
TYPICAL APPLICATIONS

HEATING OF LARGE VESSELS



- Self-adhesive film as mounting method
- Connection with silicone cap
- 2x temperature sensors in protective silicone pockets
- Custom cut-outs to fit the application

HEATING OF FILLING HOODS



- Self-adhesive film as mounting method
- Connection with silicone cap
- Temperature sensor in protective silicone pocket
- Three heating elements connected in series

TECHNICAL DETAILS

Dimensions	**max. B960 mm x L2600 mm**
Thickness	**min. 1.5 mm**
Weight	**min. 2 kg/m²**
Permitted operating temperature	**max. 200 °C**
Operating voltage	**12 - 480 V**
Heating power	**max. 0.5 W/cm²**
Ingress protection	**max. IP65 (in accordance)**
Certifications	**CE, EN45545-2, UL 94-HB**

VARIANTS OF FLAT HEATING ELEMENTS AND OPTIONS

Flat



- Optimal for heating of flat surfaces
- All kinds of cut-outs can be added
- Different mounting methods

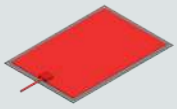


RTV adhesive



Self-adhesive film

Flat on assembly



- Flat heating element vulcanised on assembly
- Different metal plates as standard available (1 mm and 2 mm)
- Assembly can also be supplied
- No mounting methods

Flexible



- Flat heating element that can be bent (one direction can be set)
- All kinds of cut-outs can be added
- Different mounting methods



RTV adhesive



Self-adhesive film



Silicone straps



Tension springs



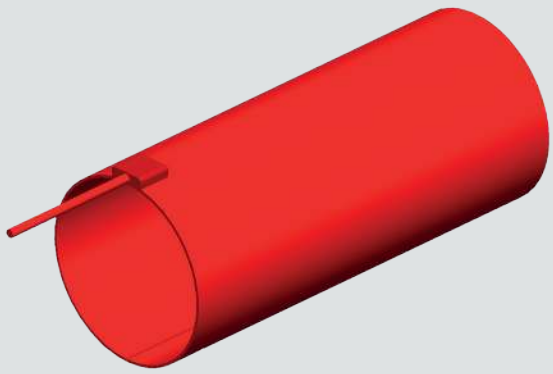
Hook and loop



Eyelets/hooks

CYLINDRICAL HEATING ELEMENTS

WOSZY



TYPICAL APPLICATIONS

HEATING THE ENDS OF SAND PIPES



- Self-adhesive film as mounting method
- Connection with reinforced silicone patch
- Temperature switch glued onto the surface
- Robust silicone connection cable

HEATING OF WATER PIPES



- Self-adhesive film as mounting method
- Connection with reinforced silicone patch
- Temperature switch glued onto the surface
- Robust silicone connection cable

TECHNICAL DETAILS

Dimensions	**max. ID310 mm x L2600 mm**
Thickness	**min. 1.5 mm**
Weight	**min. 2 kg/m²**
Permitted operating temperature	**max. 200 °C**
Operating voltage	**12 - 480 V**
Heating power	**max. 0.5 W/cm²**
Ingress protection	**max. IP65 (in accordance)**
Certifications	**CE, EN45545-2, UL 94-HB**

HEATING OF EXHAUST FILTERS



- Hook and loop as mounting method
- Connection with a silicone patch
- Insulation as a safety feature

VARIANTS OF CYLINDRICAL HEATING ELEMENTS AND OPTIONS

Cylindrical



- Optimal for heating cylindrical shapes
- All kinds of cut-outs can be added
- Different mounting methods



RTV adhesive



Self-adhesive film

Cylindrical and flexible



- Cylindrical heating element that can be bent for easy mounting and dismounting
- All kinds of cut-outs can be added
- Different mounting methods



RTV adhesive



Self-adhesive film



Silicone straps



Tension springs



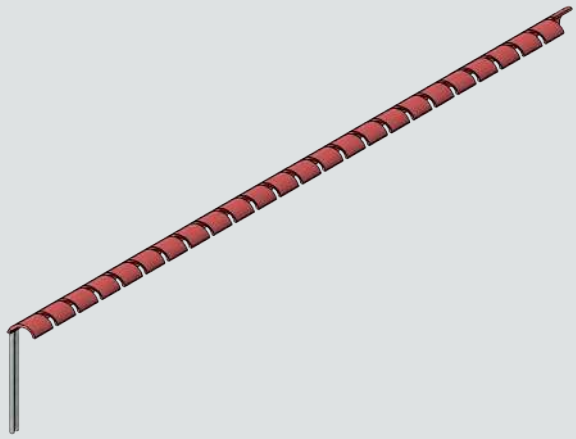
Hook and loop



Eyelets/hooks

SPIRAL HEATING ELEMENTS

WOSSP



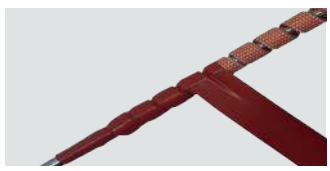
DESCRIPTION AND OPTIONS

- Flexible heating element for pipes with a small diameter
- Can be mounted on straight or curved pipes
- Mounting simply by uncoiling it on to the pipe

INSULATION



SILICONE BAND

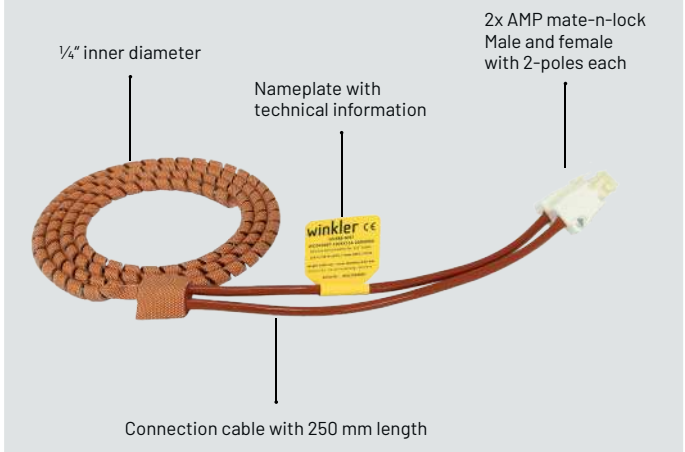


TECHNICAL DETAILS

Dimensions	**ID1/8" – 1" x L1800 mm**
Thickness	**min. 1.5 mm**
Weight	**min. 2 kg/m²**
Permitted operating temperature	**max. 200 °C**
Operating voltage	**12 - 480 V**
Heating power	**max. 0.5 W/cm²**
Ingress protection	**max. IP65 (in accordance)**
Certifications	**CE, EN45545-2, UL 94-HB**

STANDARD SPIRAL HEATERS IN STOCK

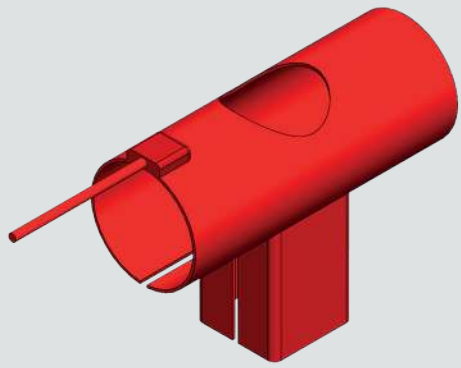
Spiral heaters with an inner diameter of ¼" and fixed connection. The standard lengths are ranging from 225 mm to 1600 mm with a heating power of approximately 0.05 W/mm when operated at 120 V. The part number for orders can be found in the table below.



Length L [mm] ± 10%	Performance P [W] @ 120 V ± 10%	Part number for order
225	11	W3072-2750
325	16	W6188-6443
425	21	W7820-2360
500	25	W3736-3624
650	32	W1278-2183
850	42	W7677-6641
1000	50	W2588-6057
1200	60	W3045-5233
1300	65	W5387-3540
1450	73	W0708-7735
1600	80	W2742-7735

FREE FORM HEATING ELEMENTS

WOSKF



TYPICAL APPLICATIONS

HEATING OF COMPLEX VALVE COMPONENTS



- No mounting method due to perfect fit
- Connection with silicone patch
- Small temperature sensor glued into the connection patch
- Silicone strands for the connection of the heating element and the temperature sensor

HEATING OF FILTER COMPONENTS



- Pressure buttons as mounting method
- Connection with reinforced silicone patch
- Temperature sensor in protective silicone pocket
- Connection cable in silicone protective sleeve

TECHNICAL DETAILS

Thickness	**min. 1.5 mm**
Weight	**min. 2 kg/m²**
Permitted operating temperature	**max. 200 °C**
Operating voltage	**12 - 480 V**
Heating power	**max. 0.5 W/cm²**
Ingress protection	**max. IP65 (in accordance)**
Certifications	**CE, EN45545-2, UL 94-HB**

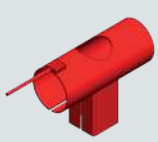
HEATING OF PUMP HOUSINGS



- Silicone jacket with inner insulation
- Hook and loops as mounting method
- Robust silicone connection cable

VARIANTS OF FREE FORM HEATING ELEMENTS AND OPTIONS

Free form



- Ideal for small complex geometries
- Complete customisation
- Also possible as insulation



RTV adhesive



Self-adhesive film



Silicone straps



Tension springs



Hook and loop



Eyelets/hooks

Free form on assembly



- Ideal for small complex geometries
- Complete customisation
- Assemblies need to be supplied

ELECTRICAL CONNECTION

CONNECTIONS ON THE SILICONE HEATING ELEMENT

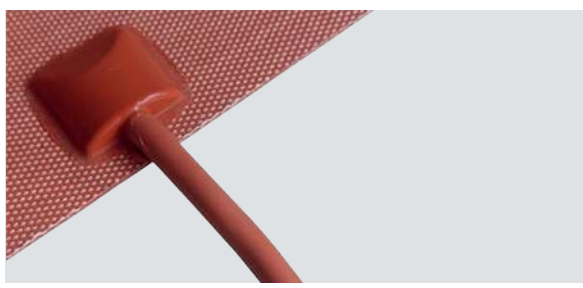
STRANDS UNDER A SILICONE COVER LAYER

Cross-section and length depending on application (for protected installation and low voltage)



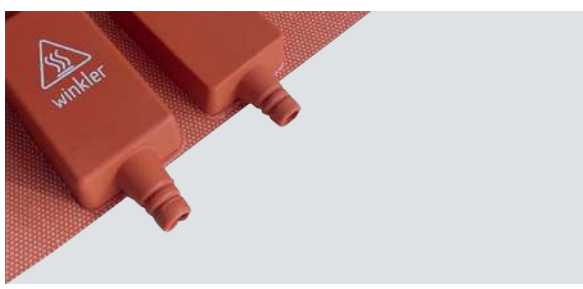
SILICONE CABLE IN SILICONE BLOCK

Two-core or multi-core, with or without protective conductor connection (if the protective conductor is looped, it is intended for earthing the heated part) possible in accordance with IP64



SILICONE CABLE IN SILICONE CAP

Two-core or multi-core, with or without protective conductor connection (if the protective conductor is looped, it is intended for earthing the heated part)



SOLDERING PADS

For complete electrical control the silicone heating elements can also be manufactured just with soldering pads.



OUR STANDARD CONNECTION CABLES



Silicone cable



Silicone-coated strands



Silicone protective sleeve

CONNECTORS



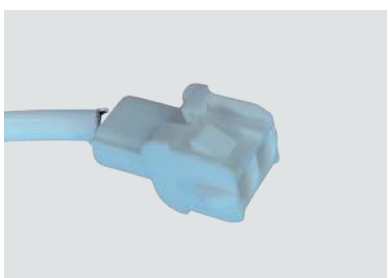
6+PE (Binder 693)



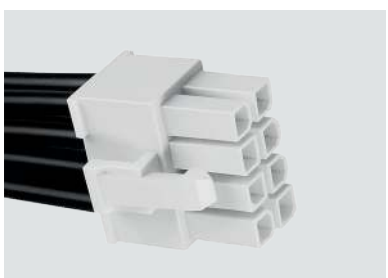
4+PE (Binder 694)



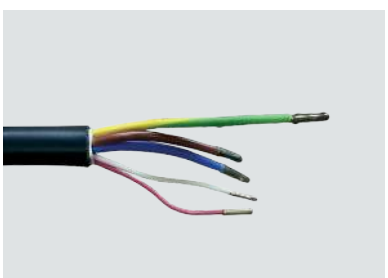
Harting 10-pole



AMP-Connector



Molex-Connector



With crimps

*Other connectors or connection cables are available on request. Alternatively connectors or connection cables can also be supplied by the customer.

TEMPERATURE CONTROL METHODS AND SAFETY

NO DESIGNATED REGULATION

You can operate your silicone heating element with no direct regulation, so the temperature is limited by the heating output and the ambient. With that option temperature fluctuations are possible.

TEMPERATURE SWITCH

Another option is to work with a temperature switch. This temperature switch, preferably a ½" disc, can be selected so that it acts as a controller in a homogeneously designed and heated environment and thus 'regulates' a steady-state heating solution.

WINKLER CONTROLLERS

Our controllers (see catalogue/website Temperature controllers) can control a heating solution in a variety of ways. This can be done by means of one or two sensors or by sensor-temperature switch-combinations.

THERMAL FUSE

In addition to the temperature switches, a thermal fuse can always be installed as an additional safety feature.

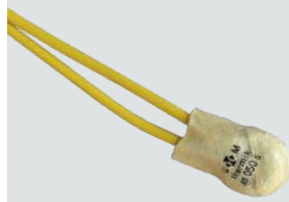




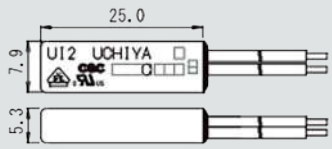
OUR STANDARD TEMPERATURE SWITCHES

Thermal protector small design
1.6 A to 42 A
50 °C - 200 °C (in 5K steps)
10 A = 10 000 switching cycles





Thermal protector flat design
2.5 A to 12 A
50 °C - 150 °C (in 5K steps)
10 A = 10 000 switching cycles
Flat design





Thermal protector can be used as limiter and controller
Up to 15 A
20 – 220 °C (in 10K steps)
10 A = 100 000 switching cycles
4 – 5 A = 500 000 switching cycles

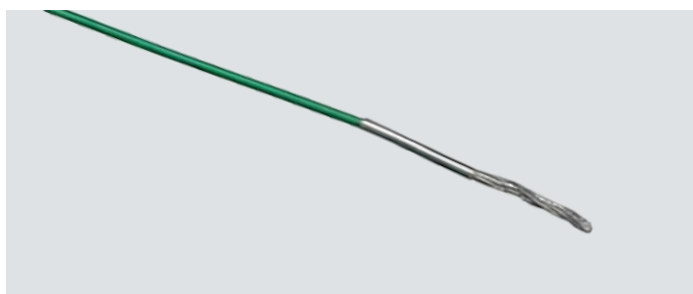


OUR STANDARD TEMPERATURE SENSORS

Sensors
Pt100
Chip Pt100
Pt1000
Type J
Type K
T/C







SILICONE HEATING PLATES



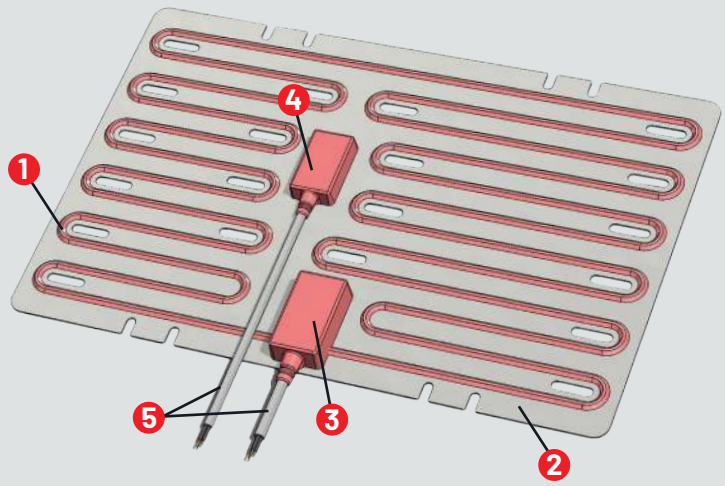
COST-EFFICIENT AND LIGHTWEIGHT HEATING PLATES FOR VARIOUS APPLICATIONS

General overview
Product examples



LIGHTWEIGHT SILICONE HEATING PLATES

WOSHE



DESIGN AND VARIANTS

1- SILICONE HEATING ELEMENT

- Meander-shaped and vulcanized silicone heating cable for uniform heat transfer

2- BASE PLATE

- Aluminium or stainless steel (can also be supplied)
- Available in thicknesses between 1 mm and 2 mm
- Optionally with mounting holes and cutouts

3- CONNECTION HEAD WITH STRAIN RELIEF

- Connection head as desired

4- TEMPERATURE SWITCH (LIMITER)

- Trigger temperatures between 50 °C and 150 °C
- Internally wired and led out

5- CONNECTION CABLE

- Lengths and customization as desired

TECHNICAL DETAILS

Permitted operating temperature	**max. 180 °C**
Heating power	**max. 0.3 W/cm²**
Operating voltage	**12–400 V**
Maximum dimensions	**1000 mm x 2600 mm**
Heater thickness (without base plate)	**3 mm**
Protection class	**I (grounded plate)**
Ingress protection	**IP65 (in accordance)**

FEATURES AND BENEFITS

- Construction with 60% less total weight
- Reduced heat-up time
- Robust for use in vehicles
- Heating of 3D geometries possible
- Free from fluoropolymers and halogens
- Conformity to EN45545-2 (in preparation)
- Reduced fire load due to less silicone

FLAT HEATING PLATES





PRODUCT EXAMPLES

3D FORM HEATING PLATES





SILICONE HEATING ELEMENTS FOR RAILWAY APPLICATIONS

winkler



LOW SMOKE LOW TOXICITY TECHNOLOGY

Railway certification overview EN 45545-2 and NFPA 130
Silicone heating elements
Silicone heating cables



RAILWAY CERTIFICATION OVERVIEW EN 45545-2 AND NFPA 130

The right heating at sensitive points plays an important role for safety, reliability, comfort and economical operation in rail technology. This applies for application in rolling stock, as in infrastructure. Time and time again, snow, ice, extreme changes in temperature and moisture cause partial or complete failure of mechanical, hydraulic, pneumatic or electrical systems. Such faults and the resulting problems in rail operation and even personal injury can be avoided with specially developed heating solutions, the appropriate controllers and thermal insulation parts.
Heating solutions from Winkler are always tailored for the respective application in coordination with the customer. Our heating solutions are space-saving and are also well suited for retrofitting. As a result of the justifiably high safety standard and the applicable industrial standards for products used in rail technology, we also continuously develop our products and meet all the relevant industrial standards. Here we are able to call upon a wide range of technologies and over 40 years of experience.

CERTIFICATIONS

Industrial standards and regulations
Fire Investigation for European Rail Transport
Certified according to **EN 45545-2**
Shock and vibration testing to **EN 61373:2010** (on request)
NFPA 130 (ASTM E162 / ASTM E662 / BSS 7239:1988)

SILICONE HEATING ELEMENTS FOR RAILWAY APPLICATIONS

EN 45545-2:2020+A1:2023
Requirement set R1 / R7 HL1; HL2
Requirement set R22 / R23 HL1; HL2; HL3

OVERVIEW

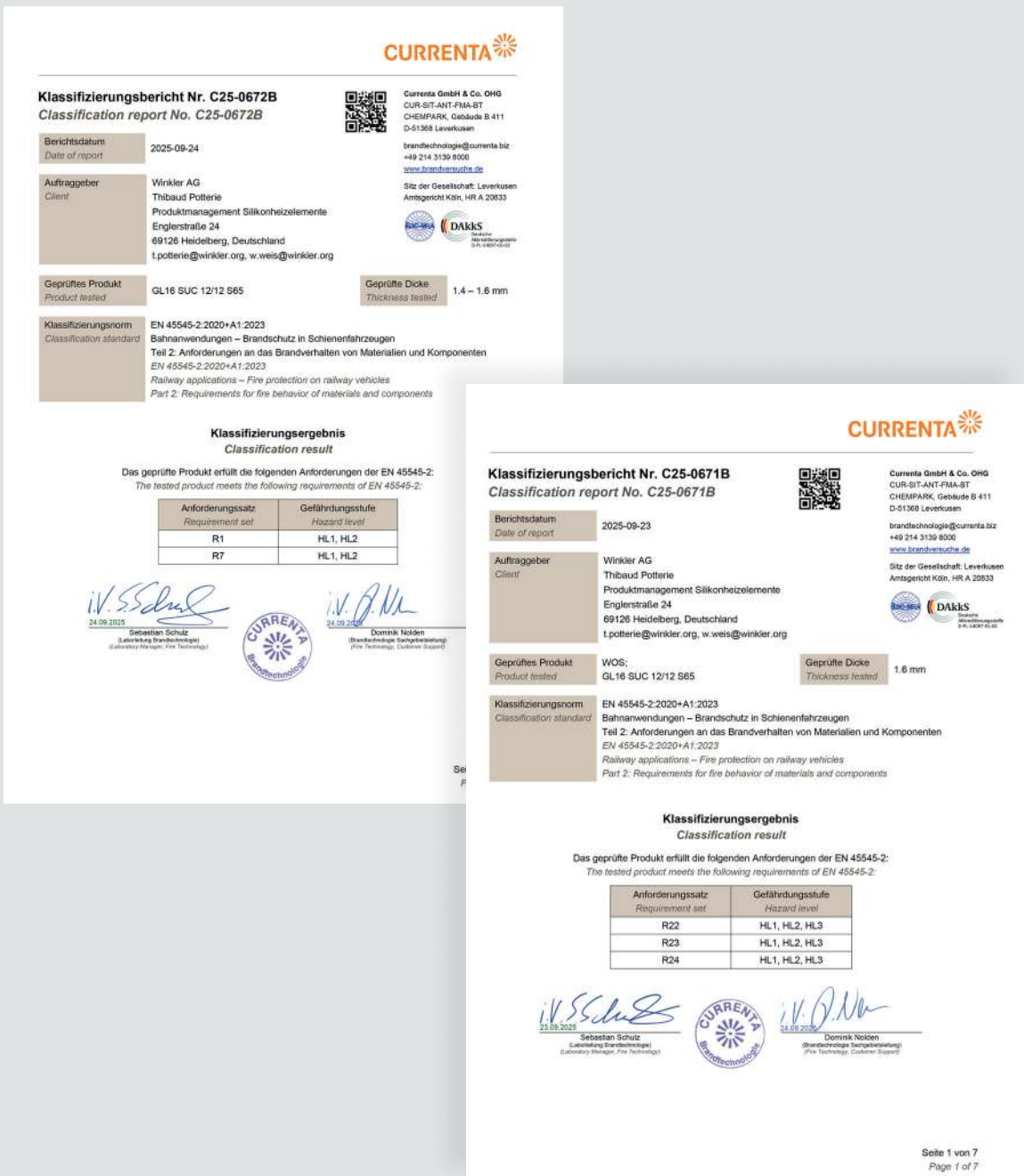


Fire test


Condition after testing

CERTIFICATES

CERTIFIED ACCORDING TO EN 45545-2



CURRENTA

Klassifizierungsbericht Nr. C25-0672B
Classification report No. C25-0672B

Currenta GmbH & Co. OHG
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CHEMPARK, Gebäude B 411
D-51368 Leverkusen

brandtechnologie@currenta.biz
+49 214 3139 8000
www.brandversuche.de

Sitz der Gesellschaft: Leverkusen
Amtsgericht Köln, HR A 20833

Berichtsdatum *Date of report*	2025-09-24
Auftraggeber *Client*	Winkler AG Thibaud Potterie Produktmanagement Silikonheizelemente Englerstraße 24 69126 Heidelberg, Deutschland t.potterie@winkler.org, w.weis@winkler.org
Geprüftes Produkt *Product tested*	GL16 SUC 12/12 S65
Geprüfte Dicke *Thickness tested*	1.4 – 1.6 mm
Klassifizierungsnorm *Classification standard*	EN 45545-2:2020+A1:2023 Bahnanwendungen – Brandschutz in Schienenfahrzeugen Teil 2: Anforderungen an das Brandverhalten von Materialien und Komponenten *EN 45545-2:2020+A1:2023* *Railway applications – Fire protection on railway vehicles* *Part 2: Requirements for fire behavior of materials and components*

Klassifizierungsergebnis
Classification result

Das geprüfte Produkt erfüllt die folgenden Anforderungen der EN 45545-2:
The tested product meets the following requirements of EN 45545-2:

Anforderungssatz *Requirement set*	Gefährdungsstufe *Hazard level*
R1	HL1, HL2
R7	HL1, HL2

24.09.2025
Sebastian Schulz
(Laborleitung Brandtechnologie)
(Laboratory Manager, Fire Technology)

24.09.2025
Dominik Nolden
(Brandtechnologie Sachgebietsleitung)
(Fire Technology, Customer Support)

CURRENTA

Klassifizierungsbericht Nr. C25-0671B
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Berichtsdatum *Date of report*	2025-09-23
Auftraggeber *Client*	Winkler AG Thibaud Potterie Produktmanagement Silikonheizelemente Englerstraße 24 69126 Heidelberg, Deutschland t.potterie@winkler.org, w.weis@winkler.org
Geprüftes Produkt *Product tested*	WOS; GL16 SUC 12/12 S65
Geprüfte Dicke *Thickness tested*	1.6 mm
Klassifizierungsnorm *Classification standard*	EN 45545-2:2020+A1:2023 Bahnanwendungen – Brandschutz in Schienenfahrzeugen Teil 2: Anforderungen an das Brandverhalten von Materialien und Komponenten *EN 45545-2:2020+A1:2023* *Railway applications – Fire protection on railway vehicles* *Part 2: Requirements for fire behavior of materials and components*

Klassifizierungsergebnis
Classification result

Das geprüfte Produkt erfüllt die folgenden Anforderungen der EN 45545-2:
The tested product meets the following requirements of EN 45545-2:

Anforderungssatz *Requirement set*	Gefährdungsstufe *Hazard level*
R22	HL1, HL2, HL3
R23	HL1, HL2, HL3
R24	HL1, HL2, HL3

23.09.2025
Sebastian Schulz
(Laborleitung Brandtechnologie)
(Laboratory Manager, Fire Technology)

24.09.2025
Dominik Nolden
(Brandtechnologie Sachgebietsleitung)
(Fire Technology, Customer Support)

Seite 1 von 7
Page 1 of 7

SILICONE HEATING ELEMENTS FOR RAILWAY APPLICATIONS

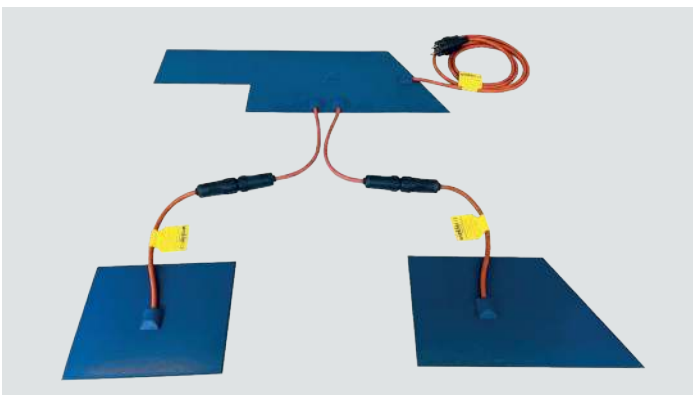
HEATING OF WATER PIPES





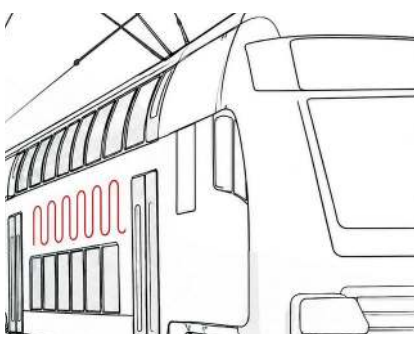
HEATING OF FOOTRESTS



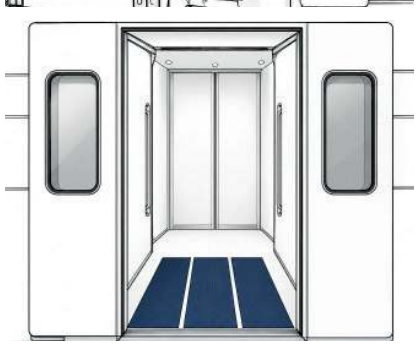


TYPICAL APPLICATION EXAMPLES

Silicone heater for heating sidewall panels



Silicone heater for heating tram entrance area



SILICONE HEATING CABLES FOR RAILWAY APPLICATIONS

winkler

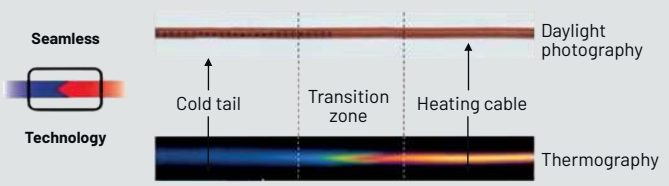
FEATURES AND BENEFITS

- **Consistently homogeneous material structure**
  Ideal performance with temperature fluctuations
- **No additional electrical connections / heat conductors / cold ends**
  Ideal behaviour with mechanical loads (especially vibrations)
- **Spiral-shaped heat conductor**
  No length variation with temperature changes
- **No change in cross-section (thickening) at transition**
  Low risk due to friction and chafing
- **No gap and no seal in the outer jacket**
  High resistance to moisture and other environmental influences

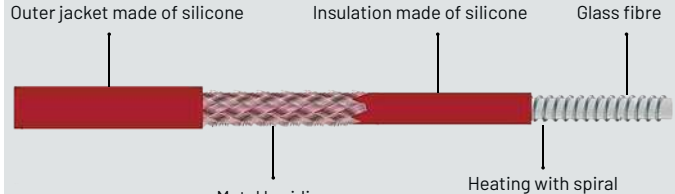
TECHNICAL DETAILS

Perm. operating temperature	**max. 150 °C**
Perm. ambient temperature	**-50...+80 °C**
Operating voltage	**max. 800 V AC/DC 2,000 V AC**
Test voltage	**2,000 V AC**
Power	**max. 40 W/m (selectable in sections)**
Ingress protection	**IP68 (in accordance)**
Length	**min. 1 m to max. 40 m (depending on electric current I max = 5A)**
Outer diameter	**5 mm**
Bending radius	**min. 25 mm**
Weight	**50 g/m**

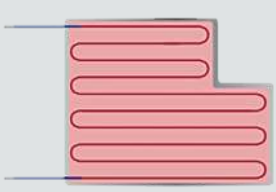
SEAMLESS TECHNOLOGY



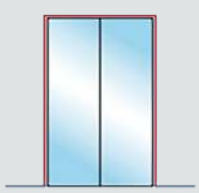
STRUCTURE



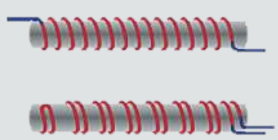
TYPICAL APPLICATION EXAMPLES



1- For heating large surfaces in floors and wall panels with meandering routing.

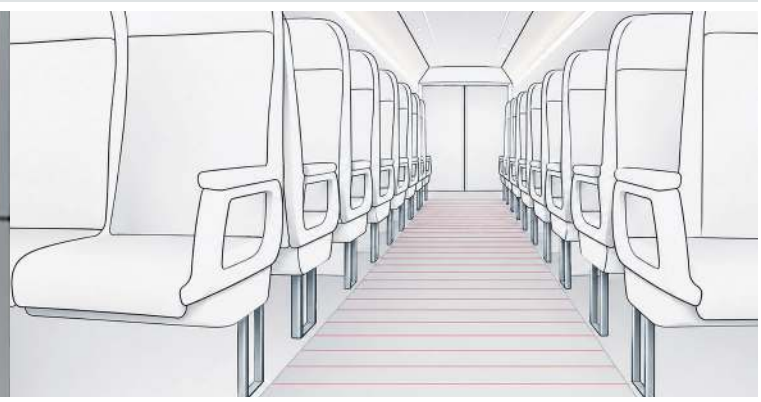


2- For heating (hollow) profiles by drawing in or longitudinal routing.



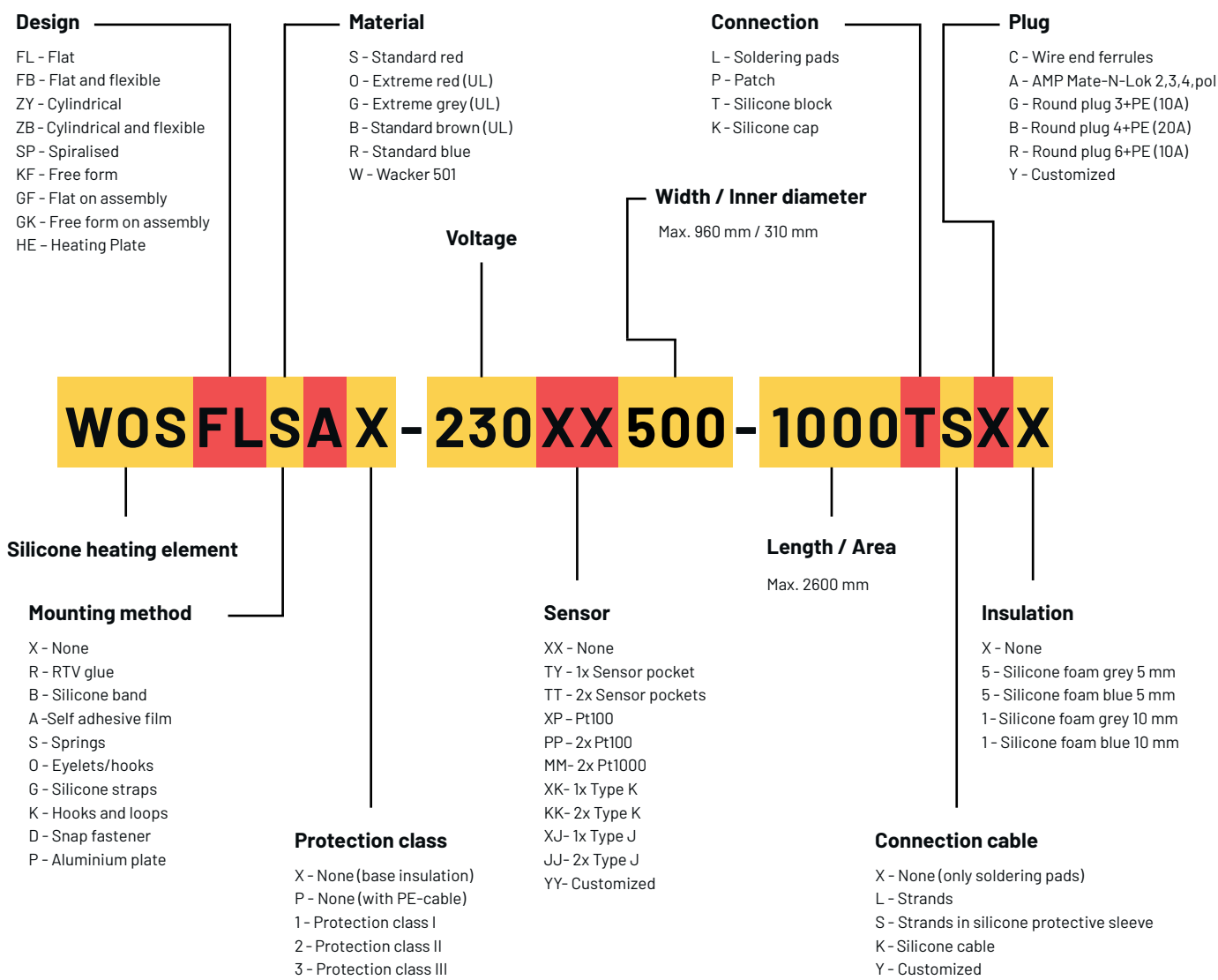
3- For heating pipes and hoses with spiral-shaped routing (0 > 50 mm) with double-sided and single-sided connection.





CLASSIFICATION OF SILICONE HEATING ELEMENTS

TYPE DESIGNATIONS



winkler

Discover our product range













READY TO HEAT.



ANY QUESTIONS? PLEASE FEEL FREE TO CONTACT US!

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